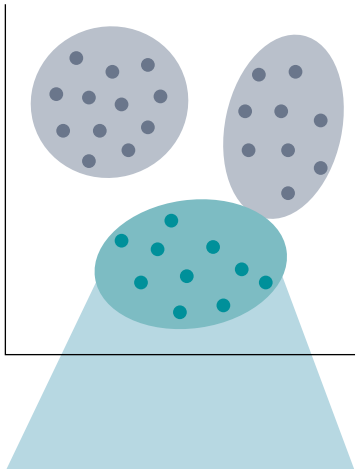
(1) Prepare single-cell multiome data



ATAC-seq fragment file

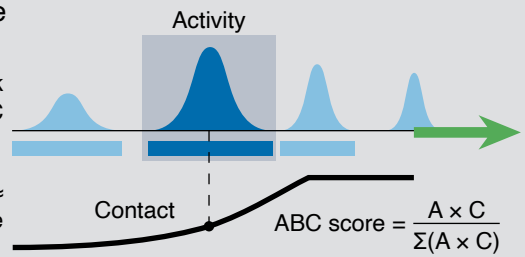
```
Chr. 1 10011   10132  CGTAGCGG
Chr. 1 10012   10151  ATGTCATC
               ...
```

RNA count matrix

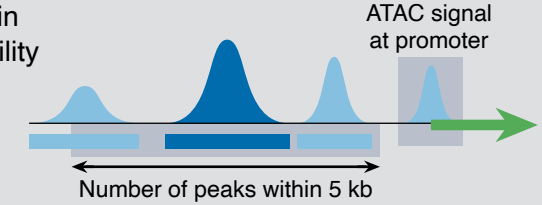
	CGTAGCGG	ATGTCATC	...
SAMD11	0	1	...
NOC2L	2	0	...
KLHL17	1	0	...
	...		

(2) Annotate peak–gene pairs

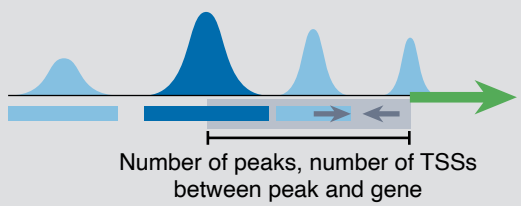
ABC score



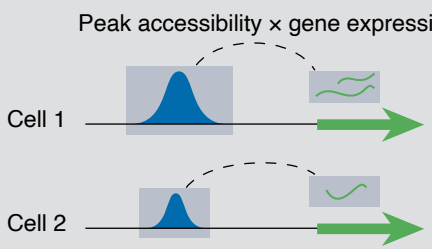
Chromatin accessibility



Gene density



Kendall correlation



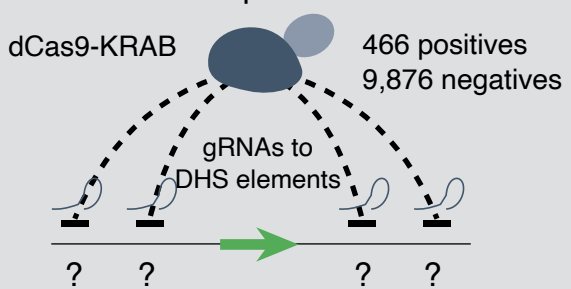
Promoter class ubiquitously expressed?

(3) Apply scE2G to make genome-wide enhancer–gene predictions

scE2G

Model training

(1) Compute features on K562 multiome data

(2) Train logistic regression classifier to predict CRISPRi-validated enhancers

CRISPR perturbation data